

STA Problem

Path ① Input to Register Period = 4 ns

$$4 > 0.5 + 2 + 0.5 \rightarrow 4 > 3 \rightarrow \text{skew} = 1 \text{ ns}$$

Path ② Reg to Reg

$$4 - \text{skew} > 0.5 + 2 + 0.5 \rightarrow \text{skew} = 1 \text{ ns}$$

Path ③ Reg to out

$$4 > 0.5 + 2 + 0.5 \rightarrow \text{skew} = 0.6$$

Path ④ In to out

$$4 > 0.5 + 1 + 0.5 \quad \text{skew} = 2$$

worst timing slack is path ③ from FF2 $\rightarrow$ out1